Stage1: Open Applications

Title of the project / product

SpikeGuard: A Speed-Adaptive, Event-Driven Bearing-Health Node for MSME Retrofit

Problem Statement:

MSME factories cannot afford always-on, speed-aware bearing monitoring on every motor, because conventional vibration nodes need continuous high-rate sampling, on-board spectral processing and connectivity that legacy machines and small plants cannot support, and fixed-frequency analysis loses sensitivity when drive speed varies

Bearing damage is often cited as roughly 30% of rotating-machinery failures, and an unplanned motor stoppage halts the line it feeds. Large plants counter this with permanently installed condition-monitoring systems. For Micro, Small and Medium Enterprises (MSMEs), per-machine sensing, wiring or gateway infrastructure and analytics subscriptions are hard to justify across dozens of ageing motors, so maintenance is often run-to-failure or based on periodic handheld checks.
Conventional vibration monitoring digitises the accelerometer continuously at kHz rates, and a processor computes FFT or envelope spectra to locate bearing defect frequencies.

This has two drawbacks for a low-power retrofit node:

First, the ADC and processor consume power at a fixed rate regardless of machine activity, and duty-cycling to save power trades away observability of intermittent faults.

Second, on variable-speed drives the defect frequencies scale with shaft speed, so fixed-frequency analysis smears fault energy across spectral bins and reduces sensitivity

Solution proposed:

SpikeGuard magnet mounted node that classifies bearing condition at the sensor. Instead of digitalizing vibration and computing FFT, it converts vibration into vectorized digital events and processes them with a spiking neural network (SNN) built in FPGA. The AI is a small network running in hardware next to the machine and will not be based on cloud computing.

How it works

• **Activity-proportional operation:** essentially no events when the machine is stopped, sparse events when healthy.

• **Speed-adaptive by construction:** hardware order tracking, with speed derived from the vibration itself, so no retrofit tachometer is needed.

• **Multiplier-free digital core** with deterministic latency.

• **Standard industrial interfaces:** 4-20 mA and Modbus. No cloud, Wi-Fi or app.

• **Simple setup:** bearing type selected from an on-node lookup table (DIP switch or Modbus register).

Implementation Plan:

Phase	Period	Stages
1. Model and design	Sep–Oct 2026	Software modelling; SPICE front-end; MATLAB speed-tracking loop; prior-art search; BOM preparation
2. Build	Oct–Nov 2026	Verilog implementation verified against the software model; motor rig with seeded faults.
3. Integrate	Dec 2026	Node on rig; experiments with the MCU baseline; initial results
4.Demonstrate	Jan–Feb 2027	Output-stage and enclosure refinement; finals demonstration; documentation

Safety and Efficiency Analysis:

Safety

• **Prevents sudden failure**. Detecting bearing degradation before seizure avoids abrupt stops, overheating and secondary damage to shafts, couplings and gearboxes.

• **Non-intrusive and advisory.** The node mounts externally and has no connection to motor power or control circuits.

• **Fail-safe signalling.** Sensor loss or an internal fault drives the loop to a defined out-of-range current so it can tell whether it's node "failure" from "healthy".

• **Electrical and mechanical safety.** Low-voltage, loop-powered design.

• **Cyber safety.** No wireless or internet exposure; data never leaves the node.

Efficiency

• **Downtime.** Turns unplanned stops into planned maintenances. The pilot will track detection lead time and faults.

• **Energy.** Energy per decision is measured against the MCU baselines rather than assumed.

• **Cost.** Removes the ADC, processor or DSP, radio and cloud subscription from the node.

• **Labour.** Continuous monitoring replaces periodic manual rounds, and no data-science skills are needed to run it.

Scalability and Future Development:

Scaling plan

• **Fleet scale.** Each node is standalone. Many nodes can share one bus gateway

• **Manufacturing scale.** Cost is concentrated in the sensor and FPGA.

• **Coverage.** The bearing lookup table grows to more bearing types, and users can enter custom orders through Modbus.

Future enhancements

• **More fault types:** gear mesh, misalignment, looseness and lubrication signatures.

• **Sensor fusion:** temperature and motor-current cross-checks.

• **Battery-free variant:** if sensor and idle power can be brought low enough, energy

harvesting could replace the loop or battery.

• **Mixed-signal ASIC:** integrate the delta modulator with subthreshold analog LIF neurons characterized in SPICE, to cut power and unit cost.

• **New environments:** pumps, fans, compressors, machine-tool spindles and gearboxes, and other event-driven sensing (acoustic emission, strain) using the same shaft-locked concept

Innovation Advantage:

SpikeGuard does not claim to be the first SNN for bearings. Its contribution is the combination:

1. a shaft-locked, order-tracking resonate-and-fire bank operating on delta-modulated envelope events, with fixed, multiplier-free neurons.

2. speed derived from the vibration itself, so legacy machines need no tachometer.

3. retrofit-grade packaging (4-20 mA, Modbus, no cloud) with a fair energy benchmark against MCU baselines.

Visual Impact & User Journey:

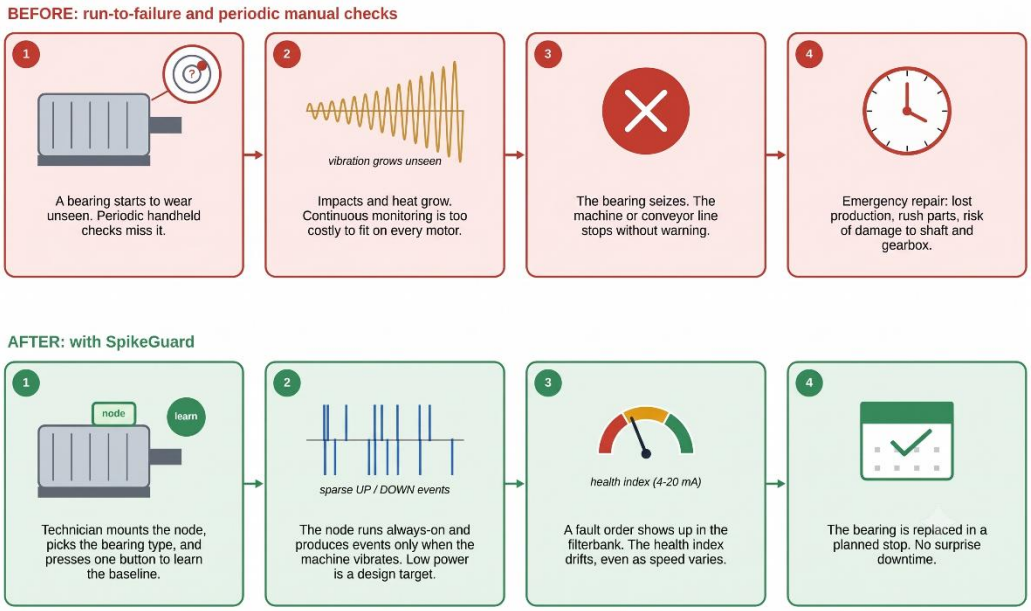


Figure. *Before-and-after user journey (illustrative storyboard).*

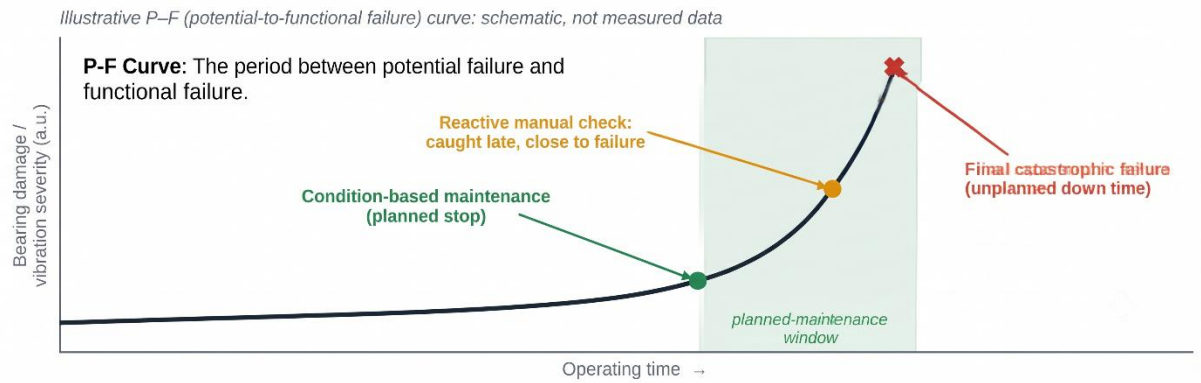


Figure. *Why early detection matters. Schematic only; the alert point is a design goal to be validated on the rig.*

Outcome for the end user: fewer emergency stops, maintenance that is planned rather than reacted to, and machine health readable from a standard 4-20 mA signal, with no new software or training.